Cumulative frequency tables and histograms

1

a 26 **b** 6 **c** 18

2

Score (x)	Frequency (f)	Cumulative frequency (cf)
18	10	10
19	9	19
20	3	22
21	7	29
Total:	29	

3

a

Number of Sightings	Number of Locations (f)	Cumulative frequency (cf)
7	4	4
8	3	7
9	5	12
10	2	14
11	5	19
12	1	20

b 6

c 14

d 9

4

a 2 **b** 12

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c

Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	1
3	2	3
4	5	8
5	5	13
6	5	18
7	9	27
8	7	34
9	5	39
10	8	47
11	1	48
12	2	50

d 7

e 11

f 18

5

Score	Frequency	Cumulative frequency
25	10	10
26	16	26
27	14	40
28	20	60
29	16	76
30	10	86

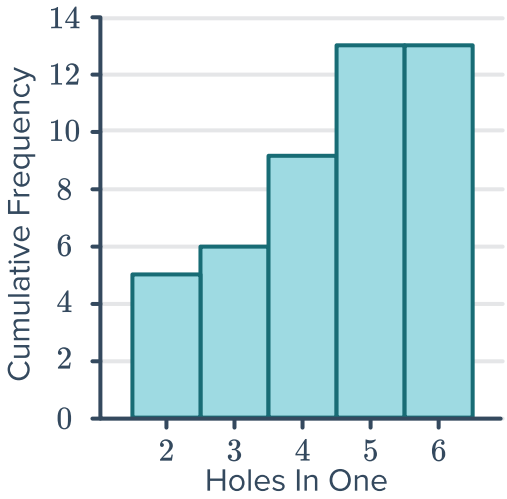
6

a 14 **b** 2 **c** 0 **d** 71.4%

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7

a



b 45

c 6

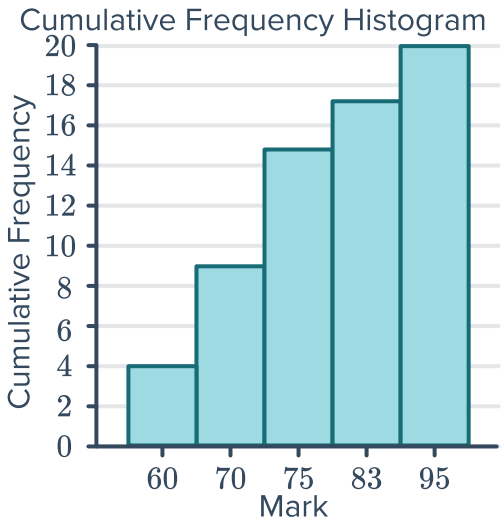
8

a The difference between the heights of two adjacent columns.

b 7

9

a



b 11

c 9

d 2

e 75

f 74.55

g 75

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10

a

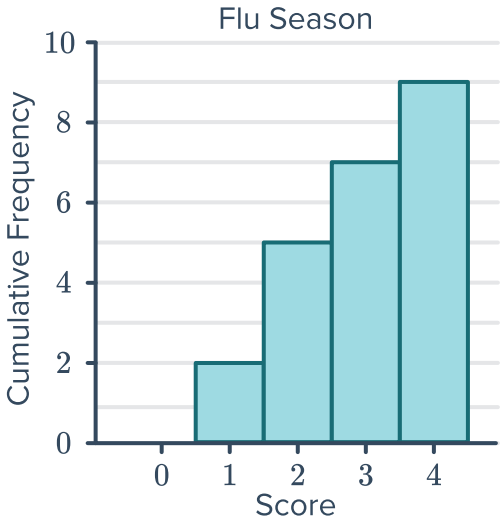
Score (x)	Frequency (f)	fx	Cumulative frequency
0	0	0	0
1	2	2	2
2	3	6	5
3	2	6	7
4	2	8	9

b 9

c 2

d 2.44

e



f 4

Cumulative frequency ogive

11

Score	Frequency
84	2
85	2
86	4
87	6
88	2

12

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13

a 30 **b** 10 **c** 10 **d** 20

e 30 **f** 20

14

a

i 12 **ii** 7 **iii** 17

b

i 675 **ii** 640 **iii** 760

c

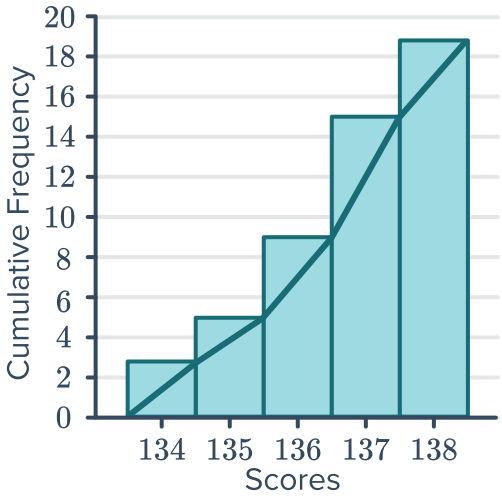
i 2100 **ii** 1900 **iii** 2450

15

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
134	3	3
135	2	5
136	4	9
137	6	15
138	4	19
Total:	19	

b



c 137

d 135

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Cumulative frequency and grouped data

16

a

i

Class	Class centre	Frequency	Cumulative frequency
$1-7$	4	11	11
$8-14$	11	14	25
$15-21$	18	10	35
$22-28$	25	15	50
$29-35$	32	24	74
Totals		74	

ii $29-35$

iii $22-28$

iv 20.55

b

i

Class	Class centre	Frequency	Cumulative frequency
$1-9$	5	8	8
$10-18$	14	16	24
$19-27$	23	4	28
$28-36$	32	21	49
$37-45$	41	16	65
Totals		65	

ii $28-36$

iii $28-36$

iv 25.91

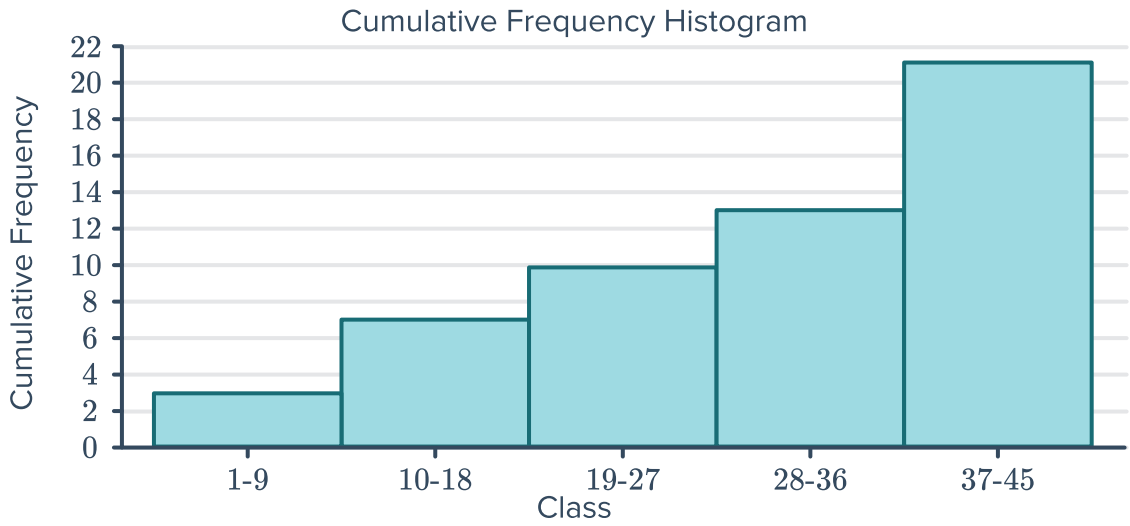
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17

a

Class	Class centre	Frequency	Cumulative frequency
$1-9$	5	3	3
$10-18$	14	4	7
$19-27$	23	3	10
$28-36$	32	3	13
$37-45$	41	8	21
Totals		21	

b



c $37-45$

d $28-36$

18

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1-4$	4	4
$5-8$	7	11
$9-12$	11	22
$13-16$	7	29
$17-20$	4	33

b 33

c 4

d Symmetrical

e 12

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19

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
$1-5$	7	7
$6-10$	15	22
$11-15$	8	30
$16-20$	11	41
$21-25$	2	43
$26-30$	1	44

b 44

c 5

d Positively skewed

e 10

20

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
$20-24$	9	9
$25-29$	18	27
$30-34$	37	64
$35-39$	14	78
$40-44$	9	87
$45-49$	6	93
$50-54$	3	96

b 96

c 5

d Positively skewed

e 34

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21

a

Class interval	Class centre(cc)	Frequency(f)	$f \times cc$	Cumul. Freq.
$51 - 55$	53	1	53	1
$56 - 60$	58	4	232	5
$61 - 65$	63	2	126	7
$66 - 70$	68	3	204	10
$71 - 75$	73	2	146	12
Total		12	761	

b $56 - 60$

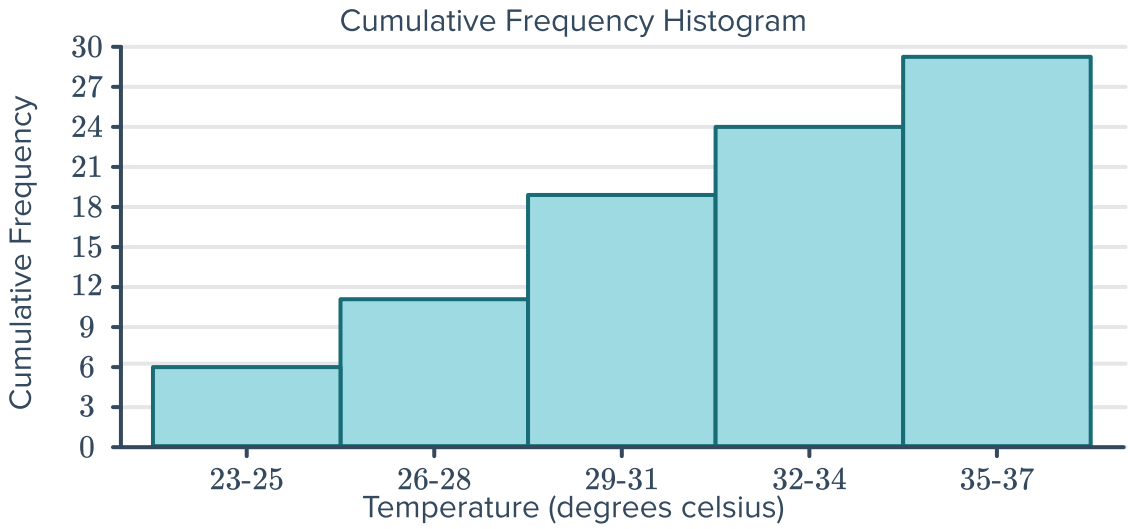
c 63.42

22

a

Class	Class centre	Frequency	Cumulative frequency
$23 - 25$	24	6	6
$26 - 28$	27	5	11
$29 - 31$	30	8	19
$32 - 34$	33	5	24
$35 - 37$	36	5	29
Totals:		29	

b



c $29 - 31^{\circ}\text{C}$

d 29.79°C

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23

a

Time Recorded	Frequency	Cumulative frequency
$16:00-16:09$	4	4
$16:10-16:19$	2	6
$16:20-16:29$	6	12
$16:30-16:39$	7	19
$16:40-16:49$	5	24
$16:50-16:59$	1	25

b 25

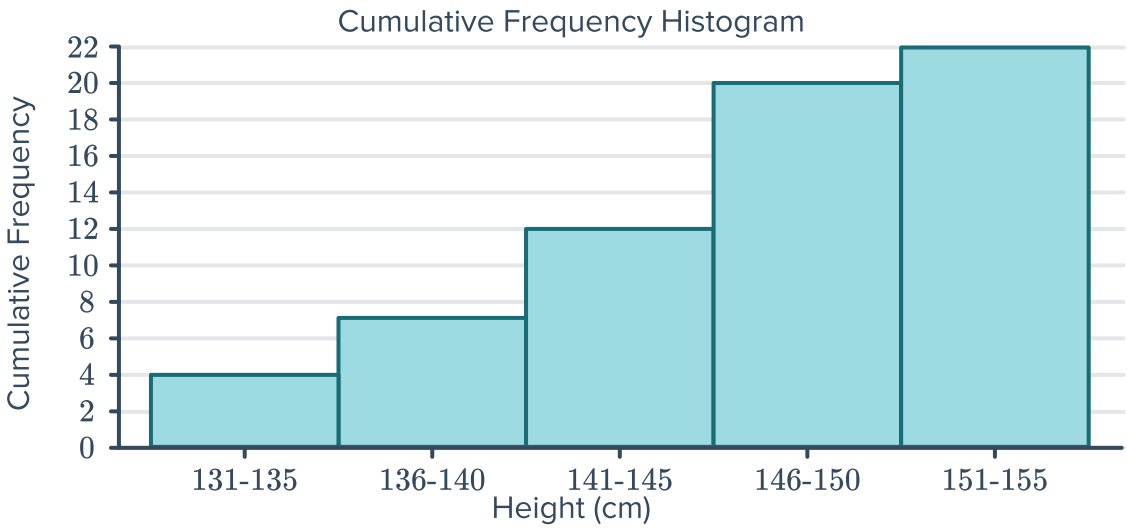
c 16:30 - 16:39

d 24

e 48%

24

a



b 2 students

c 20 students

d 5 students

e $146-150$

f 148

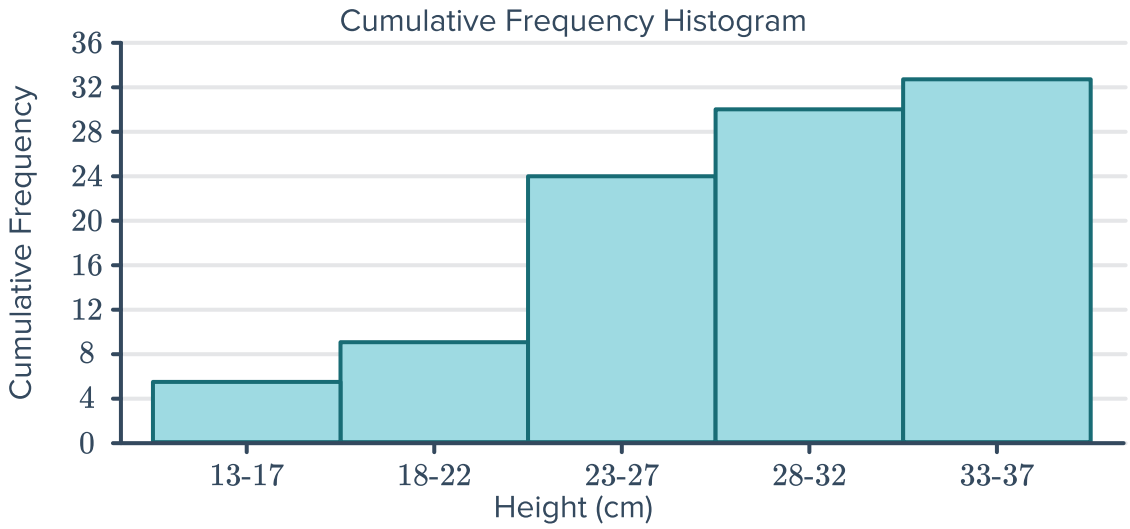
g $141-145$

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25

a 21 cm

b



c 24.55 cm

d $23 - 27$

e $23 - 27$

f 72.7%

g 27.3%

26

a

i 0.13 seconds

ii 0.132 seconds

b

Reaction Time (seconds)	Class Centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.09$	0.05	61	61
$0.09 \leq t < 0.17$	0.13	70	131
$0.17 \leq t < 0.25$	0.21	69	200

c

i 0.13 seconds

ii 0.133 seconds

d 0.758%

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